

Overview of 3D representation of the logit's predictions.

Gradient descent gives the outputs based on

$$\frac{\partial}{\partial (wX + b)} \{ \text{NLL} \} \quad \text{which returns} \quad \begin{cases} dw = (\hat{y} - y) \cdot X \\ db = (\hat{y} - y) \end{cases}$$

We use these product to adjust the logit parameters:

$$\text{logit} = \hat{w}X + \hat{b}, \quad w = \bar{w} + dw, \quad b = \bar{b} + db$$

which, through the logistic will give a prediction of y .

$$\hat{y} = \text{logistic}(\text{logit}) = \sigma(\hat{w}X + \hat{b}) \quad \rightsquigarrow \quad \hat{y} \begin{cases} \geq 0.5 & \text{flooded} \\ < 0.5 & \text{not flooded} \end{cases}$$

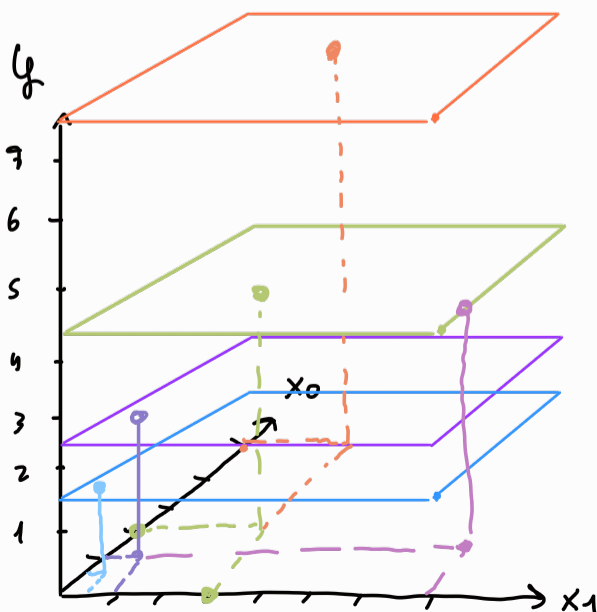
We can set $\text{logit} = 0$ which corresponds to σ

logistic exactly at 0.5.

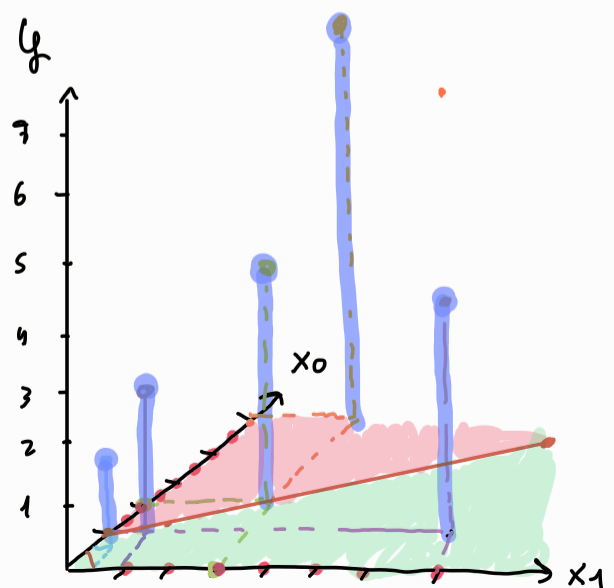
$$0 = \hat{w}_0 \cdot x_0 + \hat{w}_1 \cdot x_1 + \hat{b} \quad \rightsquigarrow \quad x_0 = - \frac{\hat{w}_1 \cdot x_1 + \hat{b}}{\hat{w}_0}$$

This gives the values of x_0 for a given range of x_1 that are at the boundary, and since it is a line equation all points follow a line.

$$X^T = \begin{bmatrix} 2 & 1 & 0.5 & 5 & 1 \\ 3 & 1 & 0.5 & 3 & 7 \end{bmatrix}, \quad \hat{w} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, \quad \hat{b} = 1$$



$$x_0 = - \frac{\hat{w}_1 \cdot x_1 + \hat{b}}{\hat{w}_0}, \quad \text{for } x_1 \in [0, \dots, T]$$



Flooded
Not flooded.